CN7023- Artificial Intelligence & Machine Vision

Week-03-Project Idea Submission

1. Topic of the Coursework Project:

   Title: Cross-View Deep Integration: A Dual-Input PyTorch Framework for Concurrent CC and MLO Mammogram Classification.

   Problem Statement: Manual evaluation of screening mammograms by radiologists has a high false-positive and false-negative rate due to structural complexity, varying breast tissue density, and incredibly subtle, low-contrast anomalies. Early and highly accurate automated breast cancer classification is essential to improve clinical workflows and survival outcomes.

2. Dataset Details: Name, Link, Size, Classes:

Dataset Name	CBIS-DDSM (Curated Breast Imaging Subset of Digital Database for Screening Mammography)
Dataset Link	CBIS-DDSM: Breast Cancer Image Dataset
Dataset Size	1.63k MB
Image Dimensions	512 x 512 pixels
Numbers of Classes	3 Pathology Classes
Number of Images per Class	Class 1: 1,367 Images; Class 2: 1,367 Images; Class 3: 400 Images
Notes	The dataset includes highly valuable, expert-validated pixel-level Region of Interest (ROI) binary masks mapping the exact boundaries of anomalies alongside full-field images. It is perfectly structured for paired-view geometric testing since it preserves matching CC and MLO configurations for patients.

3. Relevant Research Literatures using Google Scholar's to solve problems similar to this Projects and their respective links:

   a. Method 1

   - **Source Title:** Studies on deep learning approach in breast lesions detection and cancer diagnosis in mammograms
   - **Link:** https://www.utupub.fi/items/cba0cf42-0a8d-4c0b-987c-d52fd74706a4

   b. Method 2

   - **Source Title:** Breast cancer detection using artificial intelligence techniques: A systematic literature review
   - **Link:** https://www.sciencedirect.com/science/article/abs/pii/S0933365722000410

   c. Method 3

   - **Source Title:** Breast Cancer Detection Techniques: Issues and Challenges
   - **Link:** Breast Cancer Detection Techniques: Issues and Challenges | Journal of The Institution of Engineers (India): Series B | Springer Nature Link